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Roll No

CE-604 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Geo-Technical Engineering - I
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Derive the relationship between void ratio (e), degree of saturation (S_r), water content (w), and specific gravity (G) of the soil mass.
2. A soil has liquid limit and plastic limit of 47% and 33% respectively. If the volumetric shrinkage at the liquid limit and plastic limit are 44% and 29%. Determine the shrinkage limits.
3. Explain formation of soil, soil composition and structure.
4. What do you understand by consistency of soils? Explain how it is determined in laboratory?
OR
Discuss the Rebhann's graphical method for active earth pressure estimation.
5. Describe "Three phase system" of soil structure. Also define Index properties of soil mass.
6. Discuss the graphical method of determination of earth pressure on cohesive soil.
7. Describe Mohr's coulomb's theory of shear failure of soils.
Also define Newmark's influence chart.
8. Briefly explain any two of the following:
 - a) Liquefaction
 - b) Reinforced earth's retaining walls
 - c) Flownet and its uses in seepage analysis
 - d) Newmark's influence chart
 - e) Consolidation
